Ontologies, Multiagent Digital Twin and Intelligent Telemetry for Internet Infrastructure (DT-II)

Juliao Braga
UFABC S. André, SP, Brazil
0000-0001-9542-3732

Jéferson Campos Nobre
UFRGS Porto Alegre, RS, Brazil
0009-0009-0000-0000

Juliana Cristrina Braga
UFABC S. André, SP, Brazil
0000-0003-2385-0051

Itana Stiubiener
UFABC S. André, SP, Brazil
0000-0001-8166-0518

Leandro Marcio Bertholdo
UFRGS Porto Alegre, RS, Brazil
0000-0002-4700-7707

Marcelo Santos
IFSPe Salgueiro, PE, Brazil
0000-0001-5574-0645

Percival Henriques
CGI São Paulo, SP, Brazil
0009-0009-7767-9625

Lourenço Alves Pereira Júnior
ITA S. J. dos Campos, SP, Brazil
0000-0002-9682-0075

IFSPe (PE), ITA (SP), UFABC (SP), UFRGS(RS), and UNICAP (PE)

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Knowledge Base (Ontology)

The Knowledge Base

Knowledge Base (Ontology)

Description Logic (OWL)

The Knowledge Base

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Knowledge Base (Ontology)

Description Logic (OWL)

TBox

Terminological Box

ABox

Assertion Box

RBox

Role Box

The Knowledge Base

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Description Logic (OWL)

TBox

Terminological Box

Concept Definitions

- Person ⊑ Mammal[(1)]
- Car ⊑ Vehicle[(2)]
- Driver ≡ Person who drives a vehicle

ABox

Assertion Box

Facts about individuals

- Mary is a Person
- Fiat is a Car
- Mary drivex Fiat

RBox

Role Box

Relationship Rules

- drives ⊑ interactsWith[(3)]
- hasFather ⊑ hasParent
- hasBrother is symmetric[(4)]
- married is functional[(5)]

(1) Every person is a mammal
(2) Every car is a vehicle
(3) Driving is a type of interaction
(4) If A is B's brother, then B is A's brother
(5) Each person is married to at most one other person

cgi.br

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Types of Ontologies

Upper (foundational) ontology
General, domain-independent categories: object, event, quality, time… e.g. BFO, DOLCE, UFO, SUMO, GFO

guides →

Domain ontology
Concepts of a specific area, e.g. the Internet Infrastructure: AS, link, peering agreement

Why UFO?

FOUR ADVANTAGES — EVERYDAY ANALOGY → INTERNET INFRASTRUCTURE

1 What it is vs. the role it plays

A person is a student only while enrolled.

The same AS is a provider to one neighbour and a client of another, at the same time.

2 What keeps it "the same"

A football team survives a full change of players.

An AS keeps its identity as routers and links change: the RIR assignment sustains it.

3 Agreements as real things

A marriage has a start date, rules and an end.

A peering agreement has a start, terms and an end: "since when are these ASes partners?"

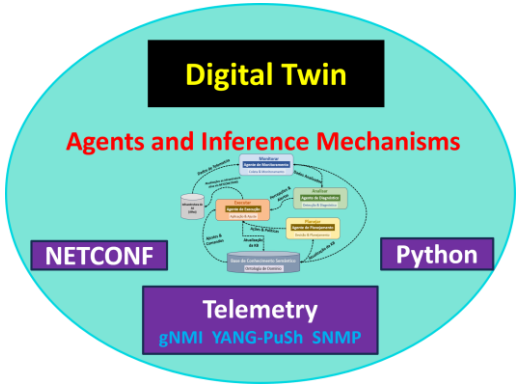
4 A common foundation

One socket standard: every appliance fits.

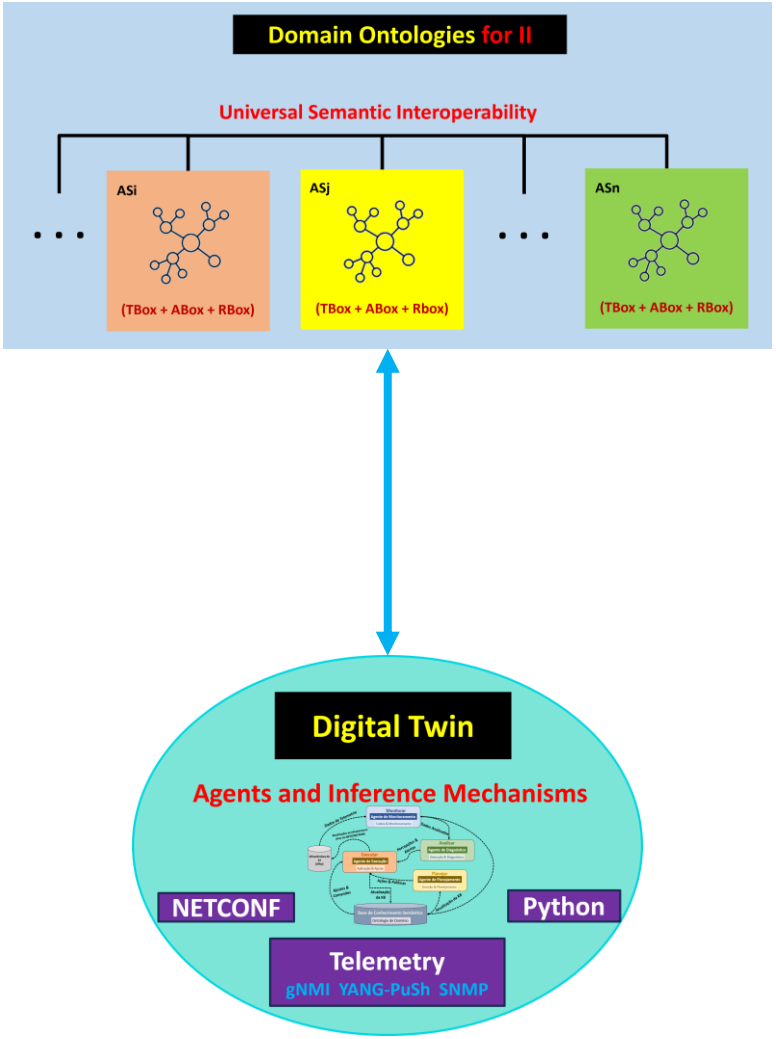
The AS ontology plugs into existing ontologies of contracts, services and risk.

Higher initial cost, fewer costly errors: worthwhile for complex, relation-rich, long-lived domains such as the Internet Infrastructure.

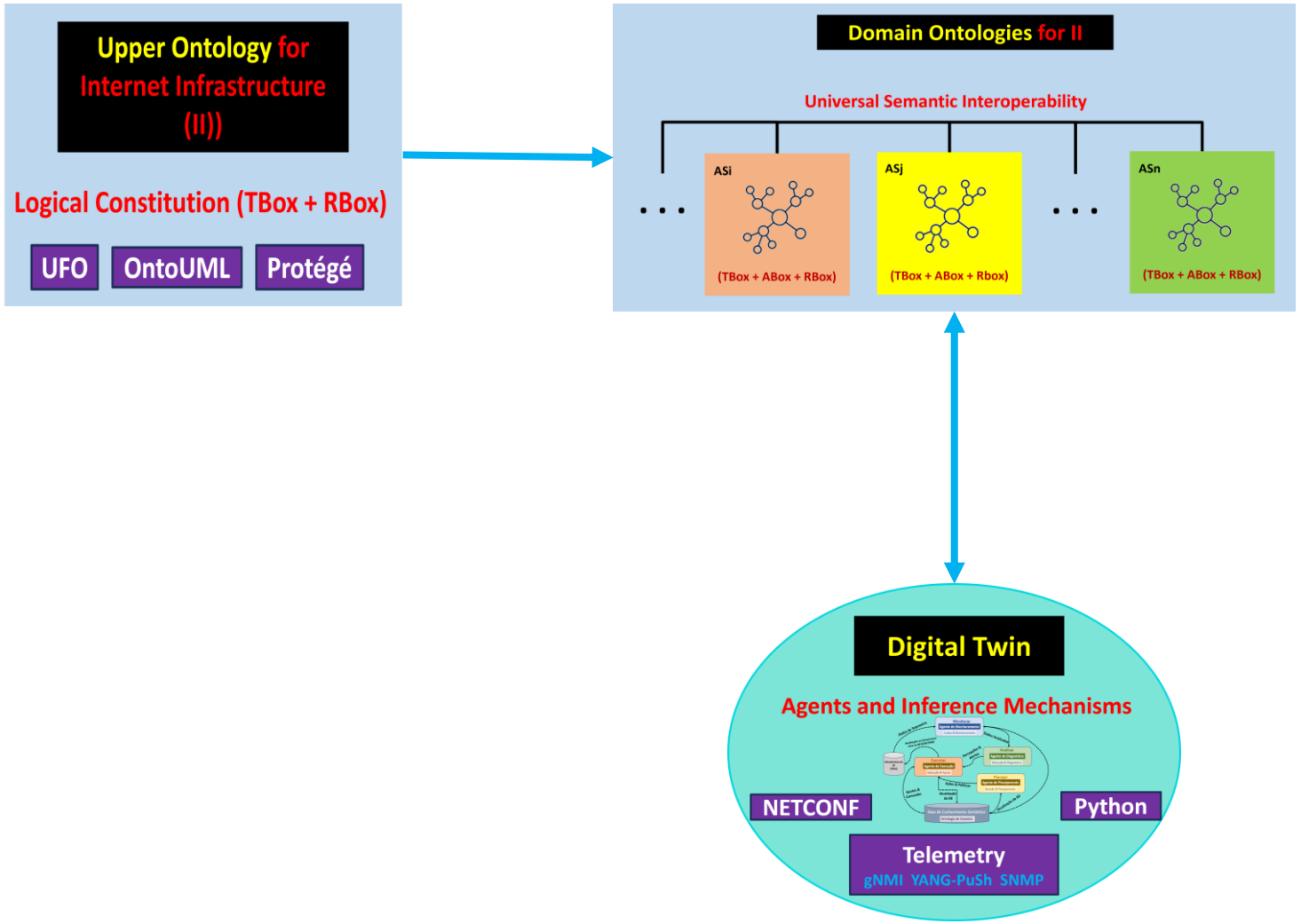
The Project


Digital Twin
Agents and Inference Mechanisms
NETCONF
Python
Telemetry
gNMI YANG-PuSh SNMP

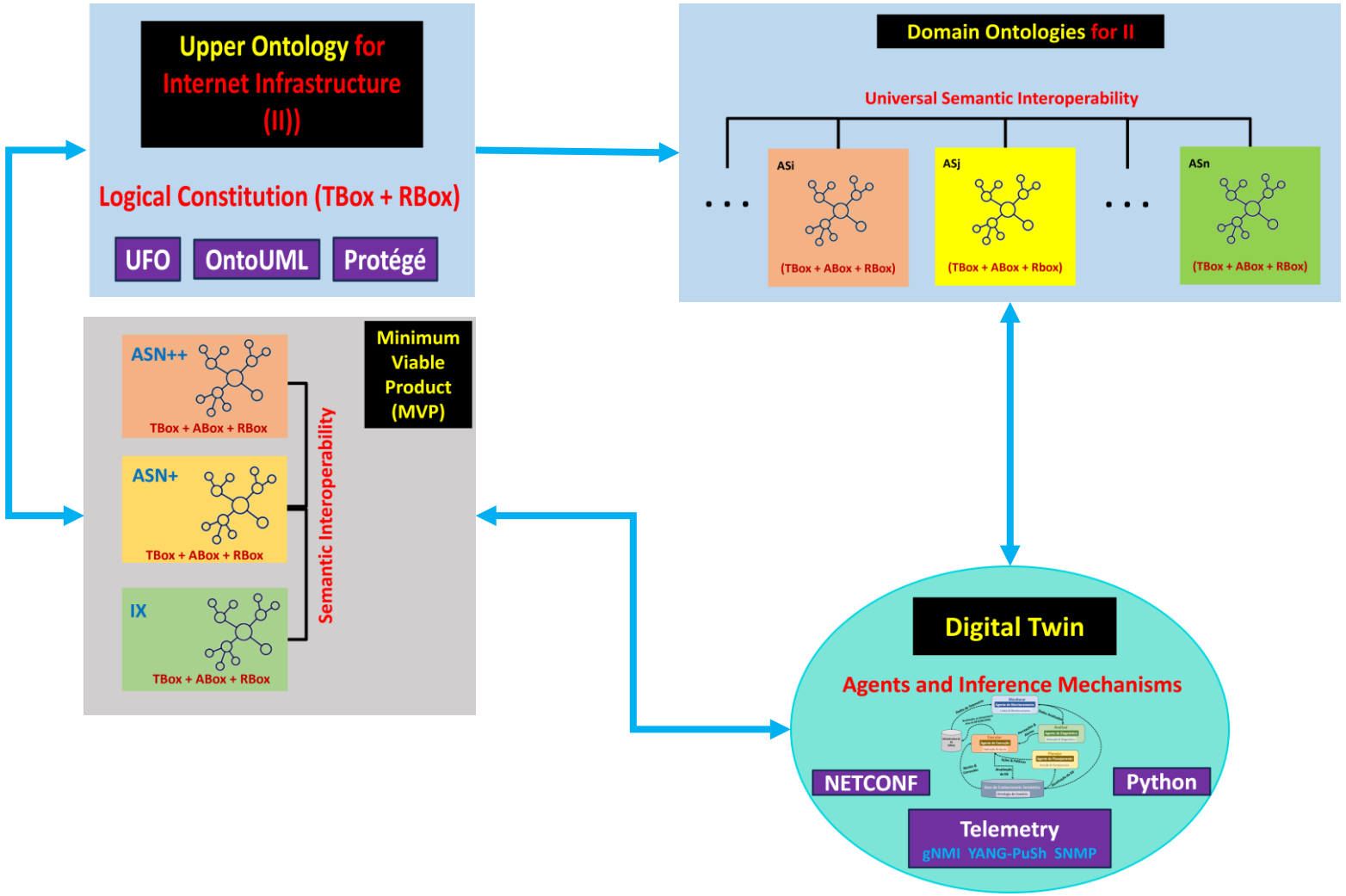
The Project


Domain Ontologies for II
Universal Semantic Interoperability
ASi
(TBox + ABox + RBox)
ASj
(TBox + ABox + Rbox)
ASn
(TBox + ABox + RBox)
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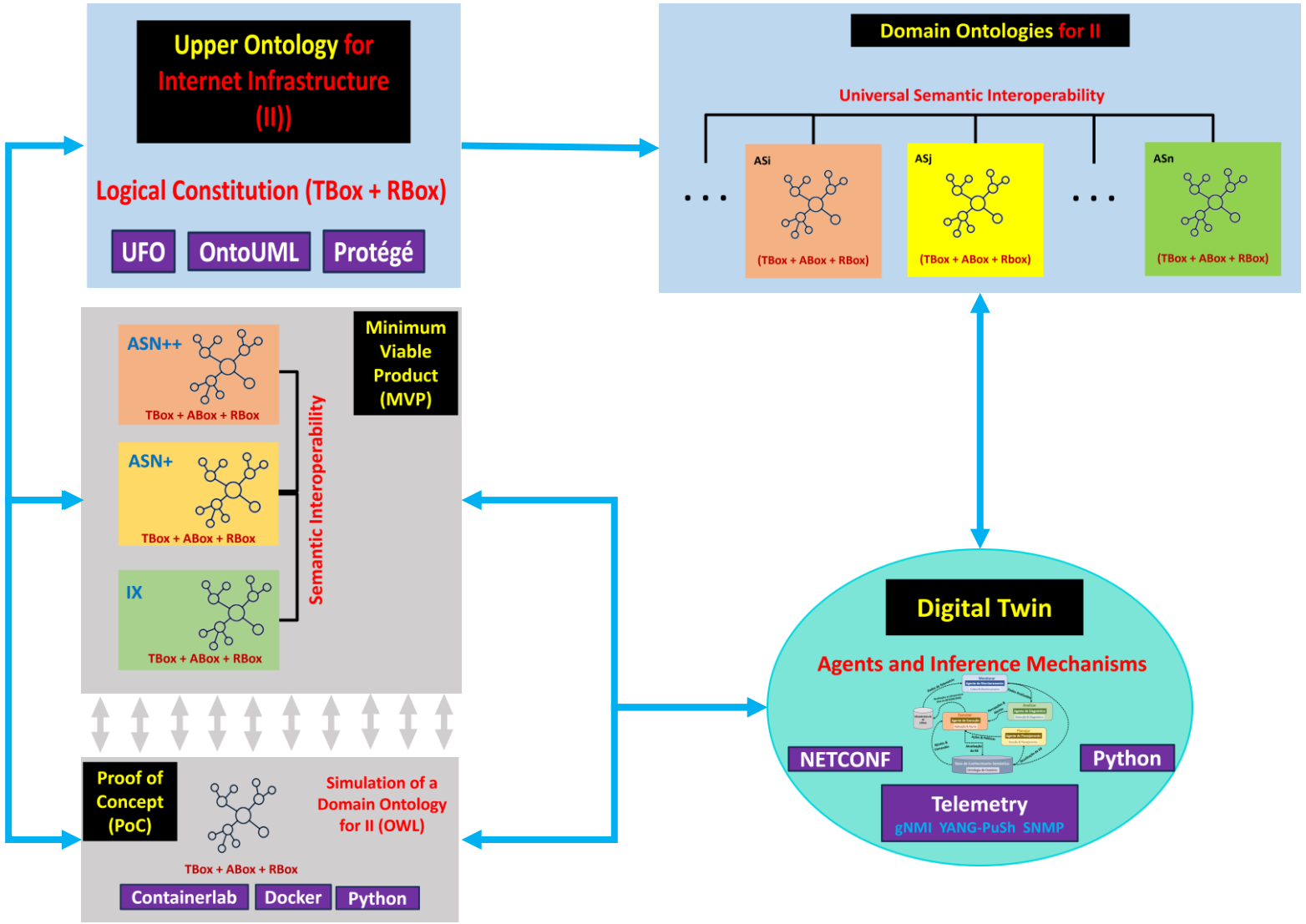
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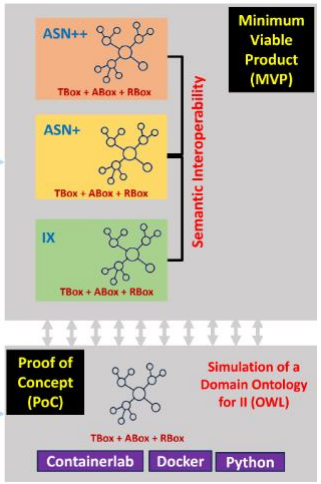

Upper Ontology for Internet Infrastructure (II))
Logical Constitution (TBox + RBox)
UFO
OntoUML
Protégé
Domain Ontologies for II
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Minimum Viable Product (MVP)
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Semantic Interoperability
Minimum Viable Product (MVP)
Proof of Concept (PoC)
Simulation of a Domain Ontology for II (OWL)
TBox + ABox + RBox
Containerlab
Docker
Python
Digital Twin
Agents and Inference Mechanisms
NETCONF
Python
Telemetry
gNMI YANG-PuSh SNMP



MVP Participants

Classification of MVP participants by level of complexity

ASN++
High complexity

PROFILE
Highly specialised, critical demands

INFRASTRUCTURE
Robust, advanced technology

SERVICES
High availability, redundancy, security

SERVES
Large corporations, public bodies, strategic sectors

TESTS IN THE ONTOLOGY
Global routes, transit policies and complex peering agreements

ASN+
Medium complexity

PROFILE
Intermediate demands, some specialisation

INFRASTRUCTURE
Adequate, less sophisticated

SERVICES
Reliable, with less redundancy

SERVES
Medium-sized companies

TESTS IN THE ONTOLOGY
The edge, where LGPD, SLAs and local access policies apply

IX
Special domain

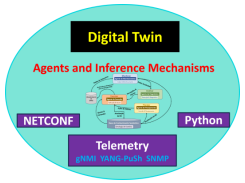
PROFILE
Neutral point serving many ASes of varying complexity

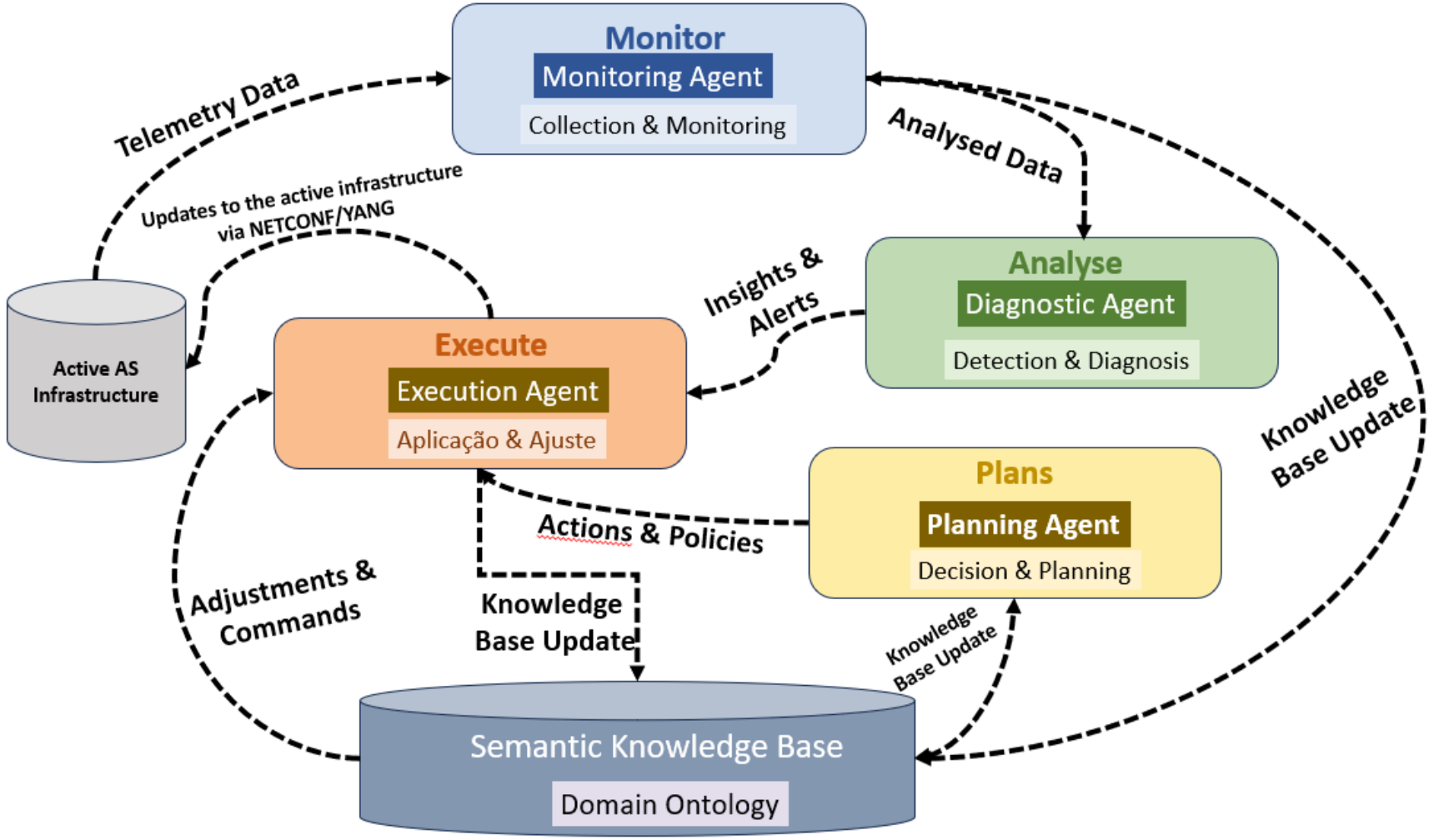
INFRASTRUCTURE
Very complex, highly diverse traffic

SERVICES
High availability, differentiated per participant

SERVES
Institutions of every size

TESTS IN THE ONTOLOGY
Relators at their limit: multilateral BGP sessions and exchange agreements





MAPE-K

Jeffrey O. Kephart and David M. Chess. “The Vision of Autonomic Computing”. In: Computer. Vol. 36. 1. IEEE,2003, pp. 41–50. doi: 10.1109/MCOM.2003.1160055

DT Reference Architecture

Ontological Layer	**Tbox: define core classes (NetworkNode, Connection, Channel) and constraints**
Identity Layer	**Assigns stable IRIs to instances, separating identity from state**
Ingestion Layer	**Collects telemetry (YANG-Push, gNMI, SNMP, syslog) and normalises data into the Node–Connection–Channel triad**
Persistence Layer	**ABox + Rbox: stores facts in RDF/OWL and maintains temporal history via observations**
Validation Layer	**Applies structural constraints and logical consistency checks with DL and DeL reasoners**
Reasoning Layer	**Infers states and propagates failures**
Exposure Layer	**Provides SPARQL queries, dashboards, alerts, and integration with NOC/orchestration systems**

Meaning-Driven Regulatory Compliance

*Jurisdiction is no longer a property of hardware but a **normative relationship** (a relator) linking a norm, the parties and the network resources.*

SCENARIO



Use Case (I)

Rule (LGPD + GDPR): health data may only traverse adequately protected jurisdictions.

T — TBox
General rules

Health-data flows only traverse adequately protected jurisdictions.

R — RBox
How relations propagate

If a node is subject to the LGPD, so are the channels and connections passing through it.

A — ABox
Facts of the moment

Flow F carries health data · F passes through node N3 · N3 is under jurisdiction J.

The Digital Twin in Action (Use Case I)

Telemetry
gNMI · YANG-Push · SNMP

→

Semantic Elevation
raw data → meaning

→

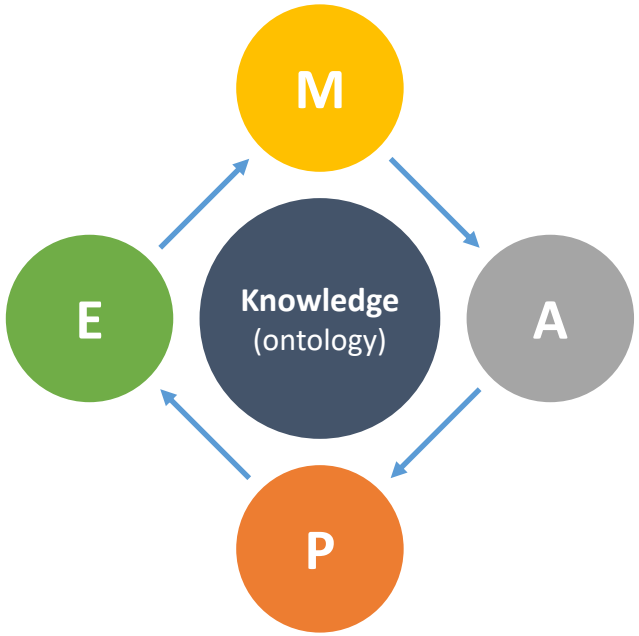
ABox
facts with meaning

→

Reasoner
checks the rules

Use Case (I)

MAPE-K LOOP AFTER A BGP FAILURE



- **M** — **Monitor:** Flow F has changed route.
- **A** — **Analyse:** F now traverses a non-compliant jurisdiction.
- **P** — **Plan:** A compliant alternative via AS-D is found.
- **E** — **Execute:** Applied via NETCONF, only after logical validation.

Takeaway: the network becomes jurisdiction-aware; compliance is checked by logical inference, not manual audits.

Responding to a DDoS Attack

*Conventional defences react to the **symptom** (suspicious IPs). DT-II reacts to the **meaning**: which services depend on what, and which agreements apply.*

Use Case (II)

SCENARIO

External botnet
attack source

→

Link_01
1 → 100 Gbps · saturated

→

DNS service
critical

→

Users
everything else depends on DNS

THE MAPE-K RESPONSE

M — Monitor
Steps 1–2

Telemetry becomes ABox facts: the flow is anomalous, Link_01 is saturated.

A — Analyse
Steps 3–4

The reasoner infers that DNS is at risk and identifies the source AS and its agreement.

P — Plan
Steps 5–7

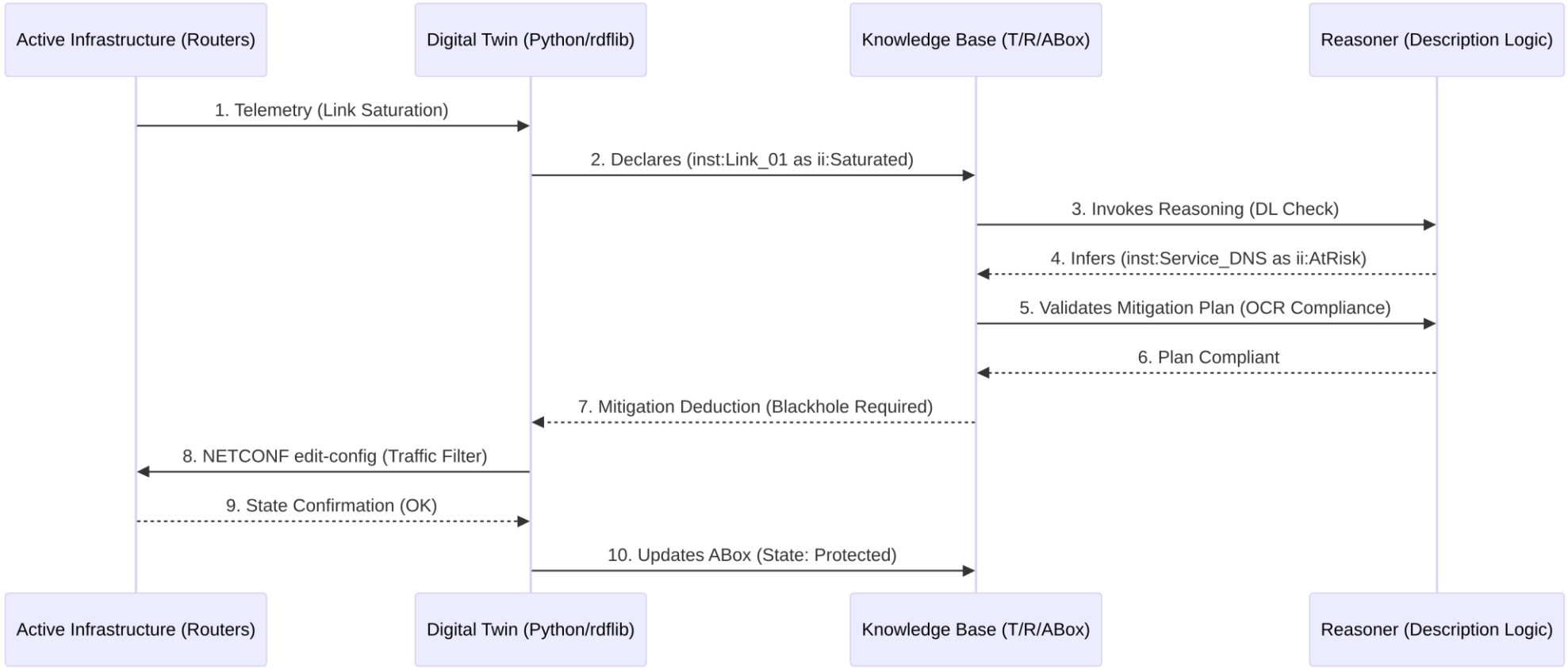
TBox rule: critical service at risk + external source → mitigate at the edge. The plan is validated first.

E — Execute
Steps 8–10

Filter applied via NETCONF, confirmed, and the ABox updated to “protected”.

Sequence Flow of the Mitigation

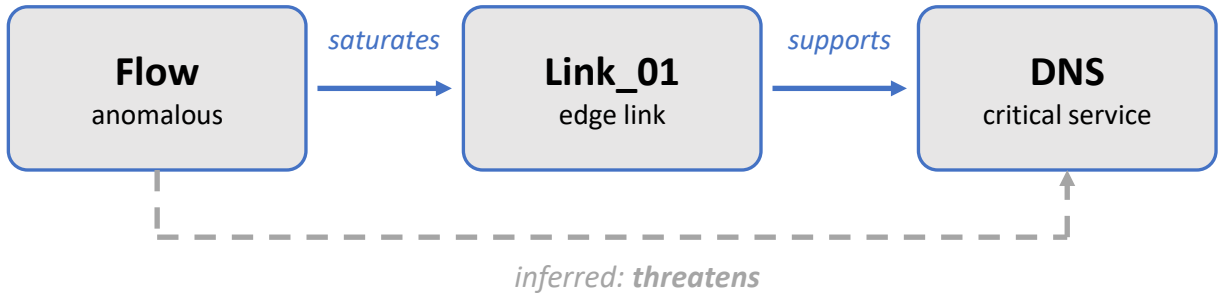
Use Case (II)



Monitor Steps 1–2 | **Analyse** Steps 3–4 | **Plan** Steps 5–7 | **Execute** Steps 8–10

Why a Semantic Defence?

HOW THE REASONER INFERS THE RISK



saturates ∘ supports ⊑ threatens

If a flow saturates a link and the link supports a service, the flow threatens the service. Nobody declares that DNS is at risk: the reasoner concludes it.

ADVANTAGES OVER IP BLOCKING

1 Context awareness

If the source AS has a cooperation agreement, ask it to filter at source instead of just blocking.

2 Mereological resilience

The model knows the parts of the network and their roles: sacrifice non-vital channels, keep DNS.

3 Traceability

Every action is explained by the axioms that justified it.

4 Compliance

No configuration is applied before it is validated against the compliance regime (Use Case I).

Use Case (II)

Takeaway: the defence knows what is at stake and why it acts, not just which IPs to block.

Contributions

1. **Reference model**
   Ontology-driven model of the Internet Infrastructure, centred on Node–Connection–Channel, grounded in UFO and built in OntoUML.

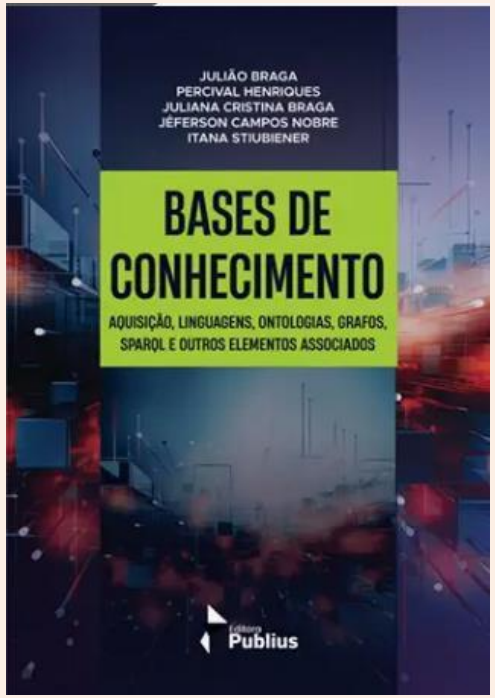
2. **Automated reasoning**
   Operationalisation in DL/OWL, with reasoning over the TBox, RBox and ABox.

3. **Live synchronisation**
   Knowledge graph kept aligned with model-driven telemetry (e.g. YANG-Push) in near real time.

4. **Governance**
   Ontological Compliance Regime and SLA constraints for auditable, explainable compliance checks.

5. **Autonomic action**
   Multi-agent Digital Twin following the MAPE-K loop for policy-driven actions.

ALSO FROM THE PROJECT



Knowledge Bases
2nd edition, now also in English, incorporating what the project learned.

Project Documentation

Project Digital Twin for Internet Infrastructure

Main: https://github.com/LJuliaoBraga/dtii

English: https://github.com/LJuliaoBraga/dtii/blob/main/docs/en-us/2026-ProjectDT-II-EN.pdf
Portuguese: https://github.com/LJuliaoBraga/dtii/blob/main/docs/pt-br/2026-ProjectDT-II-PT.pdf

ANMS 2006: https://github.com/LJuliaoBraga/dtii/blob/main/docs/events/anms%202026/AporesentacaoANMS.pdf

Thank you